

BANKING MANAGEMENT SYSTEM

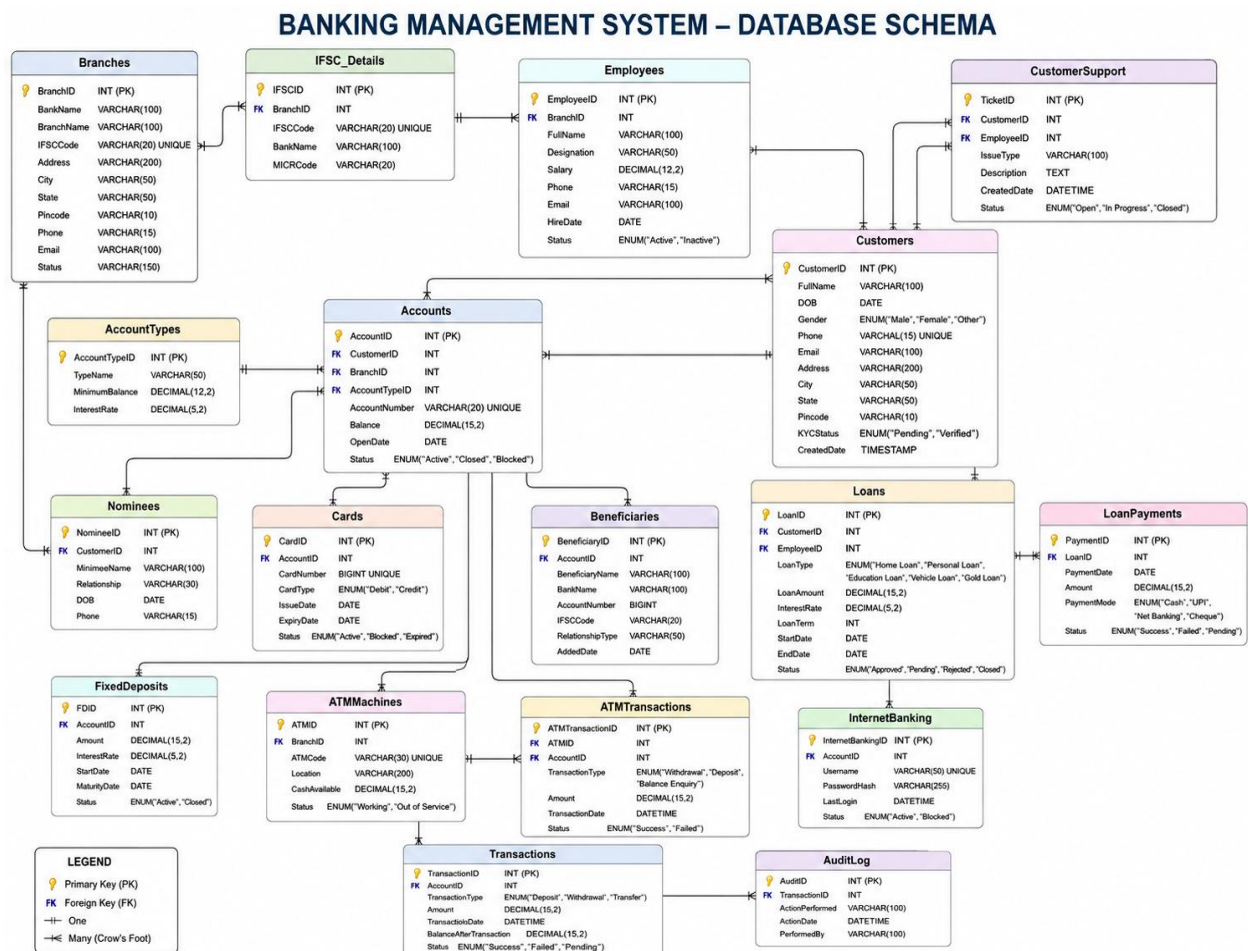
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GitHub URL: <https://github.com/yuvarani-24/BankingManagementSystem.git>

Table Schema:



BASIC LEVEL

- Write a query to find all customers whose KYC status is 'Verified'.
- Retrieve all accounts with a balance greater than ₹1,00,000.
- Find the total number of customers in each city.
- Display all employees working in each branch.
- Find the total number of accounts available in each branch.
- Retrieve customers who do not have any nominee.
- Find all loans where the loan amount exceeds ₹10,00,000.
- Display the top 10 highest account balances.
- Find the total number of ATM machines available in each branch.
- Retrieve all transactions with Status = 'Failed'.
- Find the average account balance grouped by account type.
- Retrieve employees having more than 10 years of experience.
- Find the total number of transactions performed by each account.
- Display all fixed deposits with an amount greater than ₹5,00,000.
- Find all customers having more than one bank account.
- Retrieve branches having more than 20 employees.
- Find all customer support tickets with Status = 'Open'.
- Display the latest 20 transactions based on TransactionDate.
- Find the total revenue generated through loan payments.
- Retrieve the count of accounts opened month-wise.

INTERMEDIATE LEVEL

- Display customer name, account number, branch name, and account balance.**
- Retrieve all employees along with their respective branch names.**
- Find customers who have taken loans greater than ₹5,00,000.**
- Display all ATM transactions along with ATM code and branch name.**
- Find the total account balance available in each branch.**
- Retrieve customers who have opened a Fixed Deposit.**
- Display customer names along with their debit/credit card details.**
- Find all successful transactions along with customer names.**
- Display the total loan amount issued by each branch.**
- Find the Top 5 customers having the highest account balance.**
- Retrieve customers who have both Loan and Fixed Deposit.**
- Display the total number of transactions performed by each customer.**
- Find employees who have approved more than 5 loans.**
- Display account type along with the total number of accounts.**
- Find the total amount paid for each loan.**
- Display employee names along with the number of customer support tickets handled.**
- Find customers whose account balance is below the minimum balance of their account type.**
- Display all blocked debit/credit cards with customer details.**
- Retrieve branches having more than 50 accounts.**
- Display the total amount invested in Fixed Deposits branch-wise**

ADVANCED SQL (STORED PROCEDURE, TRIGGERS)

- Detect suspicious transactions where the transaction amount exceeds the available account balance.
- Find customers who withdraw money more than 3 times within the same day.
- Detect duplicate beneficiaries having the same Account Number and IFSC Code for the same customer.
- Find customers whose balance is greater than the average balance.
- Find the second highest account balance.
- Find branches having deposits greater than the average deposit.
- Top 3 customers from every branch.
- Running balance.
- Monthly loan report using CTE.

-- Automatic Balance Update After Deposit

delimiter //

Create Trigger Depositbalanceupdate

After Insert On Transactions

For Each Row

Begin

If New.Transactiontype = 'Deposit'

And New.Status = 'Success' Then

Update Accounts

Set Balance = Balance + New.Amount

Where Accountid = New.Accountid;

End If;

End //

Delimiter ;

-- Prevent Withdrawal When Balance is Insufficient

Delimiter //

Create Trigger Preventoverdraft

Before Insert On Transactions

For Each Row

Begin

Declare Currentbalance Decimal(15,2);

Select Balance

Into Currentbalance

From Accounts

Where Accountid = New.Accountid;

If New.Transactiontype='Withdrawal'

And New.Amount > Currentbalance Then

Signal Sqlstate '45000'

Set Message_Text='Insufficient Balance';

End If;

End //

Delimiter ;

-- show procedure status like 'DIP';

-- Loan Approval Procedure

Drop Procedure If Exists Loan Approval;

Delimiter //

Create Procedure Loan approval

(

In P_Loan id Int

)

Begin

Update Loans

Set Status='Approved'

Where Loan id=P_Loan id

And Status='Pending';

End //

Delimiter ;

Call:

Call ATM Withdrawal(2,10000);

-- Duplicate Beneficiary Prevention

```
Drop Procedure If Exists AddBeneficiary;

Delimiter //

Create Procedure Addbeneficiary

(

In P_Accountid Int,

In P_Accountnumber Bigint,

In P_Ifscode Varchar(20)

)

Begin

Declare Cnt Int;

Select Count(*)

Into Cnt

From Beneficiaries

Where Accountid=P_Accountid

And

Accountnumber=P_Accountnumber

And Ifscode=P_Ifscode;

If Cnt>0 Then

Signal Sqlstate '45000'

Set Message_Text='Duplicate

Beneficiary';

Else

Select 'Beneficiary Can Be Added';

End If;

End //
```

Delimiter ;

-- **Auto Audit Log**

Delimiter //

Create Trigger Transactionaudit

After Insert On Transactions

For Each Row

Begin

Insert Into Auditlog

(

Transactionid,

Actionperformed,

Actiondate,

Performedby

)

Values

(

New.Transactionid,

'New Transaction',

Now(),

'System'

);

End //

Delimiter ;

-- Block Transaction on Closed Account

Delimiter //

Create Trigger Checkaccountstatus

Before Insert On Transactions

For Each Row

Begin

Declare Accstatus Varchar(20);

Select Status

Into Accstatus

From Accounts

Where Accountid=New.Accountid;

If Accstatus='Closed' Then

Signal Sqlstate '45000'

Set Message_Text='Account Is Closed';

End If;

End //

Delimiter ;